

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-19

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Action Center**Run date:** 2026-05-19 **Universe:** 660 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+25, 2.5%) **Clusters long:** 12/26 **ETFs long:** 370/660 (56%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (3)

cluster	theme	spread_sell	p_h5	nss
cluster_25	International Fixed Income	-0.335	0.99	-0.460
cluster_20	Inflation-Protected Fixed Income	-0.123	1.00	-0.008
cluster_10	High Yield Bond	-0.263	1.00	-0.003

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_24	Cash / Ultra-Short Fixed Income	0.362	1	1.00	1.000	1.00
cluster_21	Japan Equity	9.286	1	1.00	1.000	0.99
cluster_2	Developed ex-US Equity	3.584	1	1.00	0.772	0.33
cluster_3	U.S. Mid Cap Equity	3.488	1	1.00	0.000	0.00
cluster_14	Financials	2.153	1	0.99	0.000	-0.01

Full Sample PC1

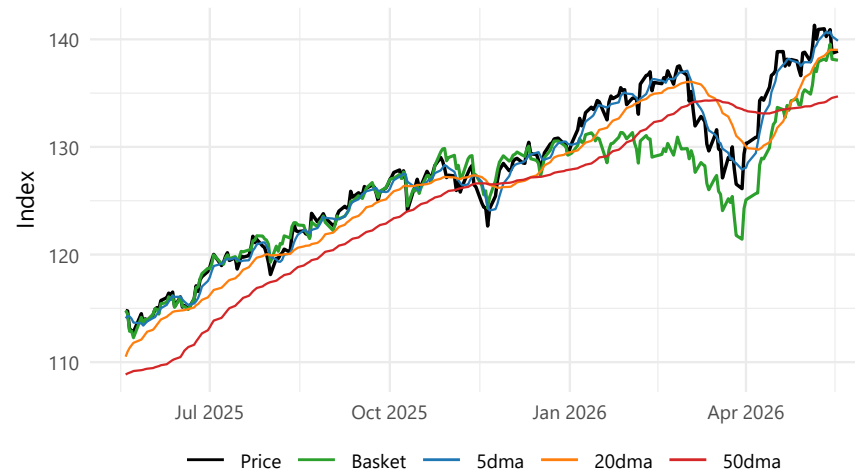


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-04-13 (25 days, 2.5%)

5dma: 139.868 20dma: 139.026 50dma: 134.697

Spread vs 50dma: +5.172 Spread vs 20dma: +0.842

P(buy crossover within 5 days): 100.0% P(sell): 63.7%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
SRLN	cluster_23	Bank Loan / Floating Rate	1.000	29	1.2%	6.23%	0.89
FLOT	cluster_23	Bank Loan / Floating Rate	1.000	29	1.2%	5.85%	0.89
QQQ	cluster_5	U.S. Large Cap Growth	1.000	26	16.3%	4.92%	0.98
BKLN	cluster_23	Bank Loan / Floating Rate	1.000	29	1.2%	4.17%	0.89
FLRN	cluster_23	Bank Loan / Floating Rate	1.000	29	1.2%	2.55%	0.89
FLTR	cluster_23	Bank Loan / Floating Rate	1.000	29	1.2%	0.69%	0.89
PDBC	cluster_22	Broad Commodities	0.983	89	51.4%	3.86%	0.98
VOO	cluster_4	U.S. Large Cap Equity	0.936	25	5.6%	7.43%	0.99
GUNR	cluster_9	Materials	0.747	29	1.8%	5.06%	0.96
AML	cluster_9	Materials	0.747	29	1.8%	0.77%	0.96
URA	cluster_9	Materials	0.747	29	1.8%	0.52%	0.96
IWP	cluster_7	U.S. Mid Cap Growth	0.739	25	6.1%	4.42%	0.98
MTUM	cluster_7	U.S. Mid Cap Growth	0.739	25	6.1%	0.94%	0.98
IEMG	cluster_8	Emerging Markets Equity	0.546	26	5.5%	4.08%	0.99
KWEB	cluster_12	China Equity	0.440	23	-1.4%	2.98%	0.92
IAU	cluster_18	Gold	0.328	4	-8.5%	1.97%	0.95
SLV	cluster_18	Gold	0.328	4	-8.5%	1.09%	0.95
GLD	cluster_18	Gold	0.328	4	-8.5%	0.07%	0.95
VTV	cluster_1	U.S. Large Cap Value	0.032	25	1.4%	10.18%	0.98
VNQ	cluster_15	Real Estate	0.027	26	2.6%	7.56%	0.99
VHT	cluster_11	Healthcare	-0.006	8	-2.1%	7.06%	0.93

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
IJH	cluster_3	U.S. Mid Cap Equity	1.00	0.000	1.046	0.99
VEA	cluster_2	Developed ex-US Equity	1.00	0.334	0.920	0.99
HYG	cluster_10	High Yield Bond	1.00	-0.003	0.916	0.95
EWJ	cluster_21	Japan Equity	1.00	0.990	0.908	0.96
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.417	0.84

SHV	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.347	0.84
MINT	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.176	0.84
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.118	0.84
VTIP	cluster_20	Inflation-Protected Fixed I...	0.99	-0.008	1.724	0.90
XLF	cluster_14	Financials	0.99	-0.008	1.097	0.96

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
5	cluster_5	U.S. Large Cap Growth	U.S. Large Cap Gr...	45	LONG		26	16.3%	1.000	-0.000
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG		29	1.2%	1.000	-0.000
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				1.000	-0.000
21	cluster_21	Japan Equity	Japan Equity	9	FLAT				0.990	-0.004
22	cluster_22	Broad Commodities	Broad Commodities	9	LONG		89	51.4%	0.983	+0.000
4	cluster_4	U.S. Large Cap Equity	U.S. Large Cap Eq...	57	LONG		25	5.6%	0.936	-0.038
9	cluster_9	Materials	Materials	21	LONG		29	1.8%	0.747	-0.179
7	cluster_7	U.S. Mid Cap Growth	U.S. Mid Cap Grow...	34	LONG		25	6.1%	0.739	-0.173
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	LONG		26	5.5%	0.546	-0.367
12	cluster_12	China Equity	China Equity	15	LONG		23	-1.4%	0.440	-0.375
2	cluster_2	Developed ex-US Equity	Developed ex-US E...	81	FLAT				0.334	-0.077
18	cluster_18	Gold	Gold	12	LONG	shifted	4	-8.5%	0.328	+1.278
1	cluster_1	U.S. Large Cap Value	U.S. Large Cap Va...	86	LONG		25	1.4%	0.032	-0.024
15	cluster_15	Real Estate	Real Estate	14	LONG		26	2.6%	0.027	-0.097
3	cluster_3	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	68	FLAT				0.000	-0.005
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT	→FLAT			-0.003	-0.014
11	cluster_11	Healthcare	Healthcare	17	LONG		8	-2.1%	-0.006	+0.025
14	cluster_14	Financials	Financials	14	FLAT				-0.008	+0.002
20	cluster_20	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT	→FLAT			-0.008	-0.029
19	cluster_19	Consumer Staples	Consumer Staples	11	FLAT				-0.353	+0.173
25	cluster_25	International Fixed Income	International Fix...	6	FLAT	→FLAT			-0.460	-0.384
17	cluster_17	Municipal Bond	Municipal Bond	13	FLAT				-0.619	-0.506
6	cluster_6	Aggregate Fixed Income	Aggregate Fixed I...	43	FLAT				-0.998	-0.036
16	cluster_16	India Equity	India Equity	14	FLAT	shifted			-0.999	-1.732
13	cluster_13	Fixed Income	Fixed Income	15	FLAT				-1.000	-0.008
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT				-1.000	+0.000

Cluster Details

Cluster 1 – U.S. Large Cap Value

Members: 86 **Var explained:** 0.875 **Signal:** LONG since 2026-04-13 (25 days, 1.4%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.3%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +17.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	23	0.2664
U.S. Large Cap Value	10	0.1223
Industrials	6	0.0688
Value Equity	4	0.0472
U.S. Low Volatility Equity	4	0.0437
U.S. Multifactor Equity	3	0.0359
U.S. Value Factor Equity	3	0.0356
U.S. Mid Cap Equity	3	0.0355

Dividend Equity	3	0.0339
Utilities	3	0.0329
U.S. Fundamental Equity	2	0.0244
U.S. Mid Cap Value	2	0.0244
U.S. Equity	2	0.0243
U.S. Equal Weight Equity	2	0.0243
U.S. Small Cap Low Volatility	2	0.0222
U.S. Mega Cap Value	1	0.0121
U.S. Revenue Weighted Equity	1	0.0120
Large Cap Blend Equities	1	0.0120
U.S. Size Factor Equity	1	0.0119
U.S. Buyback Equity	1	0.0118
U.S. Large Cap Equity	1	0.0117
U.S. Quality / Factor Equity	1	0.0114
U.S. Mid Cap Low Volatility	1	0.0114
U.S. ESG Equity	1	0.0113
U.S. Moat Equity	1	0.0113
U.S. Defensive Equity	1	0.0111
Global Equity	1	0.0107
Communication Services	1	0.0099
Real Estate	1	0.0094

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
VTV	1.027	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	RSP	0.0123	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
34	IWD	0.0123	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
37	JPUS	0.0123	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
44	VONV	0.0123	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
51	SCHV	0.0123	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
18	PRF	0.0122	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
23	CFA	0.0122	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
26	NULV	0.0122	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
30	FNDX	0.0122	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
36	IVE	0.0122	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
41	VOOV	0.0122	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG

48	ILCV	0.0122	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
50	IUSV	0.0122	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
52	SPYV	0.0122	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
54	VTV	0.0122	Vanguard Value ETF	U.S. Large Cap Value	LONG
66	IWS	0.0122	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
70	VOE	0.0122	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
75	DTD	0.0122	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
80	DGRO	0.0122	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
21	AIVL	0.0121	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
32	IWX	0.0121	iShares Russell Top 200 Value ETF	Value Equity	LONG
33	ONEV	0.0121	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
45	MGV	0.0121	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
47	VYM	0.0121	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
74	DLN	0.0121	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
85	JPME	0.0121	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
8	FEX	0.0120	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
19	RWL	0.0120	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
31	EQAL	0.0120	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
55	FTA	0.0120	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	FLAT
67	ONEY	0.0120	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
72	PFM	0.0120	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
38	SIZE	0.0119	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
49	IMCB	0.0119	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
81	VIG	0.0119	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
86	JVAL	0.0119	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
5	FVD	0.0118	First Trust VL Dividend	U.S. Dividend Equity	FLAT
7	PKW	0.0118	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	FLAT
25	NOBL	0.0118	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
27	SDY	0.0118	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	FLAT
28	FDVV	0.0118	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
35	IWR	0.0118	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
46	SCHD	0.0118	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
53	VO	0.0118	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
12	PWV	0.0117	Invesco Large Cap Value ETF	Value Equity	LONG
16	IYJ	0.0117	iShares U.S. Industrials ETF	Industrials	LONG
58	SDOG	0.0117	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
60	DVY	0.0117	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
77	DIA	0.0117	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
13	RDVY	0.0116	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
39	VLUE	0.0116	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
79	LGLV	0.0116	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	FLAT
6	FXR	0.0115	First Trust Industrials AlphaDEX	Industrials	FLAT
17	REGL	0.0115	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	FLAT
40	VIS	0.0115	Vanguard Industrials ETF	Industrials	LONG
42	FIDU	0.0115	Fidelity MSCI Industrials Index ETF	Industrials	LONG

69	SPYD	0.0115	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
73	OUSX	0.0115	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
1	XLI	0.0114	State Street Industrial Select Sector S...	Industrials	LONG
24	CFO	0.0114	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
62	RPV	0.0114	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
64	XMLV	0.0114	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
68	SMMV	0.0114	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
71	FTCS	0.0114	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	FLAT
14	MOAT	0.0113	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
20	DHS	0.0113	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
82	ETHO	0.0113	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
11	FIW	0.0112	First Trust Water ETF	Utilities	LONG
78	USMV	0.0112	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
9	PHO	0.0111	Invesco Water Resources ETF	Utilities	FLAT
22	CDC	0.0111	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
61	RDIV	0.0111	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
83	PAVE	0.0111	Global X U.S. Infrastructure Developmen...	Industrials	LONG
56	PEY	0.0110	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
63	SPHD	0.0110	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
15	FDL	0.0109	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
57	DIV	0.0108	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
65	XSLV	0.0108	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
3	ACWV	0.0107	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
29	LVHD	0.0107	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
59	SMDV	0.0107	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
10	CGW	0.0106	Invesco S&P Global Water Index ETF	Utilities	FLAT
43	HDV	0.0106	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
76	SPLV	0.0103	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
84	IYZ	0.0099	iShares U.S. Telecommunications ETF	Communication Services	FLAT
2	REM	0.0094	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 2 – Developed ex-US Equity

Members: 81 **Var explained:** 0.838 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +44.8%)
- Latest sell_cross: 2026-05-04
- Current 252d return: +23.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-04 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2721
Developed ex-US Equity	9	0.1194
International Dividend Equity	6	0.0758
Global ex-US Equity	5	0.0665
Europe Equity	4	0.0525
Alternatives	4	0.0470
Eurozone Equity	3	0.0377
Asia-Pacific Equity	2	0.0245
Real Estate	2	0.0228
Global Equity	2	0.0225
Emerging Markets Fixed Income	2	0.0195
Currency	2	0.0183
Commodity	1	0.0134
Global ex-US Small Cap Equity	1	0.0130
Germany Equity	1	0.0125
France Equity	1	0.0125
United Kingdom Equity	1	0.0124
Italy Equity	1	0.0123
Europe Financials	1	0.0122
Sweden Equity	1	0.0121
Netherlands Equity	1	0.0121
Australia Equity	1	0.0121

Spain Equity	1	0.0117
Canada Equity	1	0.0116
Switzerland Equity	1	0.0115
Belgium Equity	1	0.0115
Austria Equity	1	0.0114
Poland Equity	1	0.0105
South Africa Equity	1	0.0101
New Zealand Equity	1	0.0093
Mexico Equity	1	0.0090

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
VEA	0.92	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	VEA	0.0135	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
2	EFA	0.0134	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
49	GSIE	0.0134	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
52	ESGD	0.0134	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
63	IEFA	0.0134	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0134	Schwab International Equity ETF	Developed ex-US Equity	LONG
65	SPDW	0.0134	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
21	DWM	0.0133	WisdomTree International Equity Fund	International Dividend E...	FLAT
28	PXF	0.0133	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
34	TLTD	0.0133	FlexShares Morningstar Developed Market...	International Equity	FLAT
38	ACWX	0.0133	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
41	INTF	0.0133	iShares International Equity Factor ETF	International Equity	FLAT
59	IXUS	0.0133	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0133	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
62	VXUS	0.0133	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0132	iShares Europe ETF	Europe Equity	FLAT
35	JPIN	0.0132	JPMorgan Diversified Return Internation...	International Equity	FLAT
39	CWI	0.0132	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
42	IQLT	0.0132	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0132	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
46	RODM	0.0132	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0132	Vanguard International High Dividend Yi...	International Dividend E...	LONG
48	FNDF	0.0132	Schwab Fundamental International Equity...	International Equity	LONG

58	IEUR	0.0132	iShares Core MSCI Europe ETF	Europe Equity	FLAT
61	VGK	0.0132	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
24	IQDF	0.0131	FlexShares International Quality Divide...	International Equity	LONG
31	SCZ	0.0131	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
32	EFV	0.0131	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
33	FNDC	0.0131	Schwab Fundamental International Small ...	International Equity	LONG
56	SCHC	0.0131	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
8	DLS	0.0130	WisdomTree International SmallCap Fund	International Equity	LONG
19	EZU	0.0130	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
57	VSS	0.0130	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
29	EFG	0.0129	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
43	IVLU	0.0129	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0129	NYLI FTSE International Equity Currency...	International Equity	LONG
4	FEP	0.0128	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
45	FEZ	0.0128	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0128	Vanguard International Dividend Appreci...	International Dividend E...	LONG
30	GWX	0.0127	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0127	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
5	VIDI	0.0126	Vident International Equity Strategy ETF	International Equity	LONG
1	EWG	0.0125	iShares MSCI Germany Index Fund	Germany Equity	LONG
16	EWQ	0.0125	iShares MSCI France Index Fund	France Equity	FLAT
20	IDV	0.0125	iShares International Select Dividend E...	International Dividend E...	LONG
53	HDEF	0.0125	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
78	IPKW	0.0125	Invesco International BuyBack Achievers...	International Equity	LONG
17	EWU	0.0124	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
50	IDLV	0.0124	Invesco S&P International Developed Low...	International Equity	FLAT
22	EPP	0.0123	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
81	EWI	0.0123	iShares MSCI Italy ETF	Italy Equity	LONG
23	EUFN	0.0122	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
51	EFAV	0.0122	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
80	BBAX	0.0122	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	EWA	0.0121	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0121	State Street SPDR S&P International Div...	International Dividend E...	LONG
71	EWD	0.0121	iShares MSCI Sweden ETF	Sweden Equity	LONG
76	EWN	0.0121	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
36	DBEF	0.0120	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
11	PID	0.0119	Invesco International Dividend Achiever...	International Dividend E...	LONG
12	HEZU	0.0119	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
37	HEFA	0.0119	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
15	EWI	0.0117	iShares MSCI Spain ETF	Spain Equity	FLAT
18	EWC	0.0116	iShares MSCI Canada Index Fund	Canada Equity	LONG
55	VNQI	0.0116	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
9	HEDJ	0.0115	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0115	WisdomTree International Hedged Quality...	Alternatives	LONG
14	EWL	0.0115	iShares MSCI Switzerland ETF	Switzerland Equity	LONG

25	NFRA	0.0115	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
79	EWK	0.0115	iShares MSCI Belgium ETF	Belgium Equity	FLAT
75	EW0	0.0114	iShares MSCI Austria ETF	Austria Equity	LONG
7	RWX	0.0112	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
26	IGF	0.0110	iShares Global Infrastructure ETF	Global Equity	FLAT
70	EPOL	0.0105	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0101	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
66	EBND	0.0098	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
67	EMLC	0.0098	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
69	FXA	0.0098	Invesco CurrencyShares Australian Dolla...	Currency	LONG
73	ENZL	0.0093	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWX	0.0090	iShares MSCI Mexico ETF	Mexico Equity	LONG
77	FXC	0.0085	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT

Cluster 3 – U.S. Mid Cap Equity

Members: 68 **Var explained:** 0.906 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +41.0%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +20.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1519
U.S. Mid Cap Equity	6	0.0912
U.S. Small Cap Value	6	0.0904
U.S. Small Cap Growth	6	0.0890
Consumer Discretionary	6	0.0802
Industrials	5	0.0664
U.S. Mid Cap Value	3	0.0454
U.S. Mid Cap Growth	3	0.0445
Leveraged / Inverse	2	0.0291
U.S. Small Cap Fundamental Equity	1	0.0153
U.S. Small Cap ESG Equity	1	0.0153
Unknown	1	0.0153
U.S. Small/Mid Cap Fundamental Equity	1	0.0153
U.S. Mid/Small Cap Equity	1	0.0153
Mid Cap Value Equities	1	0.0152
U.S. Small Cap Multifactor Equity	1	0.0152
Small Cap Value Equities	1	0.0150
Dividend Equity	1	0.0150
U.S. Small/Mid Cap Equity	1	0.0149
U.S. Extended Market Equity	1	0.0149
U.S. Mid Cap Dividend Equity	1	0.0148
U.S. Mid Cap Quality Equity	1	0.0147

Value Equity	1	0.0147
Growth Equity	1	0.0147
U.S. Equity	1	0.0147
U.S. Small Cap Dividend Equity	1	0.0146
U.S. Small Cap Momentum Equity	1	0.0144
U.S. Small Cap Industrials	1	0.0144
U.S. Micro Cap Equity	1	0.0142
U.S. Small Cap Consumer Discretionary	1	0.0139

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
IJH	1.046	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
6	FNX	0.0153	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
9	NUSC	0.0153	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
10	PRFZ	0.0153	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
13	MDY	0.0153	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
21	IVOO	0.0153	Vanguard S&P Mid-Cap 400 ETF		LONG
22	VI00	0.0153	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
25	IJR	0.0153	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
28	IJH	0.0153	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
29	SPMD	0.0153	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
30	SPSM	0.0153	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
31	VB	0.0153	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
32	SCHA	0.0153	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
46	FNDA	0.0153	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
55	VBR	0.0153	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
3	IWM	0.0152	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
5	SCHM	0.0152	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
7	FYX	0.0152	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
11	SMLF	0.0152	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
14	IJT	0.0152	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
16	VIOG	0.0152	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
19	SLYG	0.0152	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
23	VTWO	0.0152	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
42	EZM	0.0152	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
51	MDYV	0.0152	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT

4	JHMM	0.0151	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
47	IWN	0.0151	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
48	IJJ	0.0151	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
50	IVOV	0.0151	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
54	VTWV	0.0151	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
61	RWK	0.0151	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
41	EES	0.0150	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
43	PSC	0.0150	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
49	IJS	0.0150	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
52	SLYV	0.0150	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
53	VIOV	0.0150	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
67	SDVY	0.0150	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT
15	IJK	0.0149	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
18	MDYG	0.0149	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
27	VXF	0.0149	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
56	RWJ	0.0149	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
68	ISMD	0.0149	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
17	IVOG	0.0148	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
40	DON	0.0148	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
26	ISCG	0.0147	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
58	XMHQ	0.0147	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	LONG
60	RFV	0.0147	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
62	OMFS	0.0147	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
66	FYT	0.0147	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
12	IWO	0.0146	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
39	DES	0.0146	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
20	VTWG	0.0145	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
24	VBK	0.0144	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
33	FXD	0.0144	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
57	PSCI	0.0144	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
63	RZV	0.0144	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
65	XSMO	0.0144	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
8	IWC	0.0142	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
38	RSPD	0.0142	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
35	FTXR	0.0139	First Trust Nasdaq Transportation ETF	Industrials	LONG
64	PSCD	0.0139	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
45	XTN	0.0137	State Street SPDR S&P Transportation ETF	Industrials	FLAT
1	XRT	0.0135	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
59	PKB	0.0135	Invesco Building & Construction ETF	Industrials	FLAT
34	PEJ	0.0132	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
37	IYT	0.0132	iShares U.S. Transportation ETF	Industrials	FLAT
44	XHB	0.0130	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
2	JETS	0.0121	U.S. Global Jets ETF	Industrials	LONG
36	ITB	0.0119	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 4 – U.S. Large Cap Equity

Members: 57 **Var explained:** 0.906 **Signal:** LONG since 2026-04-13 (25 days, 5.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.2%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +21.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1653
U.S. Total Market Equity	7	0.1288
Global Equity	7	0.1264
Consumer Discretionary	6	0.0970
Asset Allocation	4	0.0670
U.S. Mega Cap Equity	3	0.0542
U.S. Quality / Factor Equity	3	0.0535
U.S. Multifactor Equity	3	0.0534
Communication Services	3	0.0483
U.S. ESG Equity	2	0.0366
U.S. Equity	2	0.0366

Commodity	1	0.0183
Large Cap Value Equities	1	0.0182
Large Cap Blend Equities	1	0.0182
Dividend Equity	1	0.0178
U.S. Dividend Equity	1	0.0175
Alternatives	1	0.0158
International IPO Equity	1	0.0141
Singapore Equity	1	0.0129

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
VOO	0.958	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	SPY	0.0184	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
30	IWV	0.0184	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
31	IYY	0.0184	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
36	ESGU	0.0184	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
37	IWB	0.0184	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
42	VTHR	0.0184	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
45	VONE	0.0184	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
47	VV	0.0184	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
48	ILCB	0.0184	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
49	ITOT	0.0184	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
50	IVV	0.0184	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
51	SCHB	0.0184	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
52	SCHX	0.0184	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
53	SPTM	0.0184	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
54	VOO	0.0184	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
55	VTI	0.0184	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
12	URTH	0.0183	iShares MSCI World ETF	Global Equity	LONG
26	JHML	0.0183	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
29	SUSA	0.0183	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
34	SPYX	0.0183	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
38	IWL	0.0183	iShares Russell Top 200 ETF	U.S. Equity	LONG
43	GSLC	0.0183	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
46	MGC	0.0183	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0182	iShares MSCI ACWI ETF	Global Equity	LONG

15	VT	0.0182	Vanguard Total World Stock Index ETF	Global Equity	LONG
22	TILT	0.0182	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
28	DSI	0.0182	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
32	LRGF	0.0182	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
44	EPS	0.0182	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
13	CRBN	0.0181	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
14	SPGM	0.0181	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
33	OEF	0.0181	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
39	QUAL	0.0181	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
8	AOA	0.0180	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
40	QUS	0.0179	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
23	I00	0.0178	iShares Global 100 ETF	Global Equity	LONG
24	QDF	0.0178	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
25	CATH	0.0178	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	XLG	0.0178	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
41	SPHQ	0.0176	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0175	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
27	DGRW	0.0175	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0172	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
57	OMFL	0.0171	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
18	VCR	0.0168	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
21	FDIS	0.0168	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
1	XLY	0.0166	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
10	AOM	0.0164	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0162	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0162	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0158	State Street Communication Services Sel...	Communication Services	LONG
5	QAI	0.0158	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
56	RTH	0.0156	VanEck Retail ETF	Consumer Discretionary	FLAT
9	AOK	0.0150	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
16	FPXI	0.0141	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0140	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
6	EWS	0.0129	iShares MSCI Singapore ETF	Singapore Equity	LONG

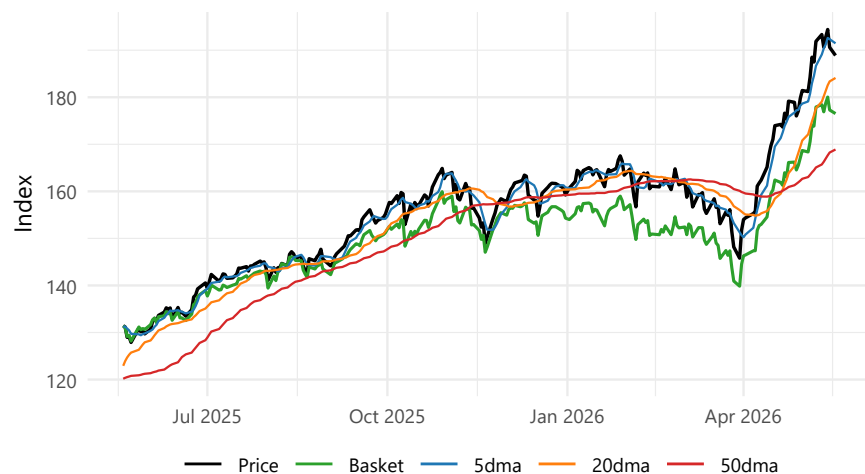
Cluster 5 – U.S. Large Cap Growth

Members: 45 **Var explained:** 0.879 **Signal:** LONG since 2026-04-10 (26 days, 16.3%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +55.4%)
- Latest sell_cross: 2026-03-06

- Current 252d return: +44.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Technology	23	0.5109
U.S. Large Cap Growth	11	0.2551
Growth Equity	2	0.0457
U.S. Equity	2	0.0454
U.S. Mega Cap Growth	1	0.0230
U.S. Equal Weight Equity	1	0.0226
Options Income	1	0.0209
U.S. Small Cap Technology	1	0.0209
Trend-Following Equity	1	0.0204
U.S. Trend-Following Equity	1	0.0178
Israel Equity	1	0.0172

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.021

ticker	rep_wt	name	category	etf_signal
QQQ	1.005	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	QQQ	0.0234	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
21	IGM	0.0234	iShares Expanded Tech Sector ETF	Technology	LONG
25	ONEQ	0.0234	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
29	VGT	0.0234	Vanguard Information Tech ETF	Technology	LONG
31	FTEC	0.0234	Fidelity MSCI Information Technology In...	Technology	LONG
23	IYW	0.0233	iShares U.S. Technology ETF	Technology	LONG
34	ILCG	0.0233	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
1	XLK	0.0232	State Street Technology Select Sector S...	Technology	LONG
22	IXN	0.0232	iShares Global Tech ETF	Technology	LONG
27	IWF	0.0232	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
32	VONG	0.0232	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
35	IUSG	0.0232	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
38	VUG	0.0232	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
17	XNTK	0.0231	State Street SPDR NYSE Technology ETF	Technology	LONG
24	NULG	0.0231	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
28	IVW	0.0231	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
30	VOOG	0.0231	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
36	SCHG	0.0231	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
37	SPYG	0.0231	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
11	QTEC	0.0230	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
26	IWY	0.0230	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
33	MGK	0.0230	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
6	AIQ	0.0228	Global X Artificial Intelligence & Tech...	Technology	LONG
16	FXL	0.0228	First Trust Technology AlphaDEX	Technology	LONG
20	PWB	0.0227	Invesco Large Cap Growth ETF	Growth Equity	LONG
10	QQEW	0.0226	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
14	XT	0.0224	iShares Future Exponential Technologies...	Technology	LONG
13	TDIV	0.0222	First Trust NASDAQ Technology Dividend ...	Technology	LONG
39	SPHB	0.0221	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
3	SMH	0.0220	VanEck Semiconductor ETF	Technology	LONG
2	SOXX	0.0219	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
45	QTUM	0.0218	Defiance Quantum ETF	Technology	LONG
5	ROBO	0.0217	ROBO Global Robotics and Automation Ind...	Technology	LONG
4	BOTZ	0.0216	Global X Robotics & Artificial Intellig...	Technology	LONG
12	PSI	0.0214	Invesco Semiconductors ETF	Technology	LONG
15	XSD	0.0214	State Street SPDR S&P Semiconductor ETF	Technology	LONG
43	FTXL	0.0214	First Trust Nasdaq Semiconductor ETF	Technology	LONG
8	NXTG	0.0213	First Trust Indxx NextG ETF	Technology	LONG
44	PTF	0.0211	Invesco Dorsey Wright Technology Moment...	Technology	LONG
19	QYLD	0.0209	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
42	PSCT	0.0209	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG

18	PTNQ	0.0204	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
9	PRNT	0.0192	3D Printing (The) ETF	Technology	LONG
40	PTLC	0.0178	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
41	EIS	0.0172	iShares MSCI Israel ETF	Israel Equity	LONG

Cluster 6 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-08 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-08)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3081
Aggregate Fixed Income	6	0.1478

Investment Grade Corporate Bond	6	0.1440
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0667
Target Maturity Bond	2	0.0422
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0226

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
BND	0.728	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
17	BND	0.0248	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
6	GVI	0.0246	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	FLAT
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
7	AGGY	0.0239	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
4	GTO	0.0238	Invesco Total Return Bond ETF	Active Fixed Income	FLAT

3	BOND	0.0237	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
38	SHY	0.0228	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
28	AGZ	0.0226	iShares Agency Bond ETF	Agency Fixed Income	LONG
18	TOTL	0.0225	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
41	VGSH	0.0225	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0220	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0214	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
8	IBDR	0.0208	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
21	LDUR	0.0171	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 7 – U.S. Mid Cap Growth

Members: 34 **Var explained:** 0.779 **Signal:** LONG since 2026-04-13 (25 days, 6.1%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.6%)
- Latest sell_cross: 2026-03-12
- Current 252d return: +17.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-12)



Category breakdown

category	n_members	abs_weight
Technology	12	0.3474
Industrials	4	0.1050
U.S. Mid Cap Growth	3	0.0973
U.S. Momentum Equity	3	0.0882
IPO Equity	2	0.0617
Convertible Bond	2	0.0615
Momentum Equity	2	0.0602
Financials	2	0.0594
Growth Equity	1	0.0313
U.S. Mid Cap Momentum Equity	1	0.0294
U.S. Small Cap Momentum Equity	1	0.0293
Consumer Discretionary	1	0.0292

Replication basket – MTUM+IWP (cor=0.98) basket lead at lag 1: 0.025

ticker	rep_wt	name	category	etf_signal
MTUM	0.183	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
IWP	0.860	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
22	IWP	0.0325	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
25	VOT	0.0324	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
26	IMCG	0.0324	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
18	FPX	0.0317	First Trust US Equity Opportunities ETF	IPO Equity	LONG
16	FTC	0.0316	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
20	RPG	0.0313	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
17	PDP	0.0311	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
19	CWB	0.0311	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
32	VFMO	0.0307	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
7	ARKQ	0.0304	ARK Autonomous Technology & Robotics ETF	Technology	LONG
23	ICVT	0.0304	iShares Convertible Bond ETF	Convertible Bond	LONG
15	FINX	0.0303	Global X FinTech ETF	Financials	FLAT
2	SKYY	0.0300	First Trust Cloud Computing ETF	Technology	LONG
11	IPO	0.0300	Renaissance IPO ETF	IPO Equity	LONG
13	FDN	0.0300	First Trust DJ Internet Index Fund	Technology	LONG
12	PNQI	0.0296	Invesco Nasdaq Internet ETF	Technology	FLAT
6	ARKW	0.0295	ARK Next Generation Internet ETF	Technology	LONG
5	ARKK	0.0294	ARK Innovation ETF	Technology	LONG
21	XMMO	0.0294	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
24	MTUM	0.0294	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
27	DWAS	0.0293	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
9	IBUY	0.0292	Amplify Online Retail ETF	Consumer Discretionary	FLAT
14	IPAY	0.0291	Amplify Digital Payments ETF	Financials	FLAT
1	IGV	0.0290	iShares Expanded Tech-Software Sector E...	Technology	LONG
8	FV	0.0290	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
10	CIBR	0.0284	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
4	HACK	0.0283	Amplify Cybersecurity ETF	Technology	LONG
34	PRN	0.0283	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
31	QMOM	0.0281	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
3	BLOK	0.0272	Amplify Blockchain Technology ETF	Technology	LONG
30	XAR	0.0268	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
28	PPA	0.0257	Invesco Aerospace & Defense ETF	Industrials	FLAT
29	ITA	0.0241	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT
33	BLCN	0.0241	Siren NexGen Economy ETF	Technology	LONG

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.875 **Signal:** LONG since 2026-04-10 (26 days, 5.5%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +38.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0812
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0358
Global Dividend Equity	1	0.0347
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.025

ticker	rep_wt	name	category	etf_signal
IEMG	0.929	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0391	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0386	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0358	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0347	Global X SuperDividend ETF	Global Dividend Equity	FLAT
25	FEMS	0.0346	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – Materials

Members: 21 **Var explained:** 0.707 **Signal:** LONG since 2026-04-07 (29 days, 1.8%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.3%)
- Latest sell_cross: 2026-03-09
- Current 252d return: +40.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-03-09)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6525
MLP / Energy Infrastructure	4	0.1709
Energy	1	0.0501
Consumer Staples	1	0.0493
Norway Equity	1	0.0439
Industrial Metals	1	0.0332

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.043

ticker	rep_wt	name	category	etf_signal
AMLP	0.117	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.080	Global X Uranium ETF	Materials	FLAT
GUNR	0.772	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
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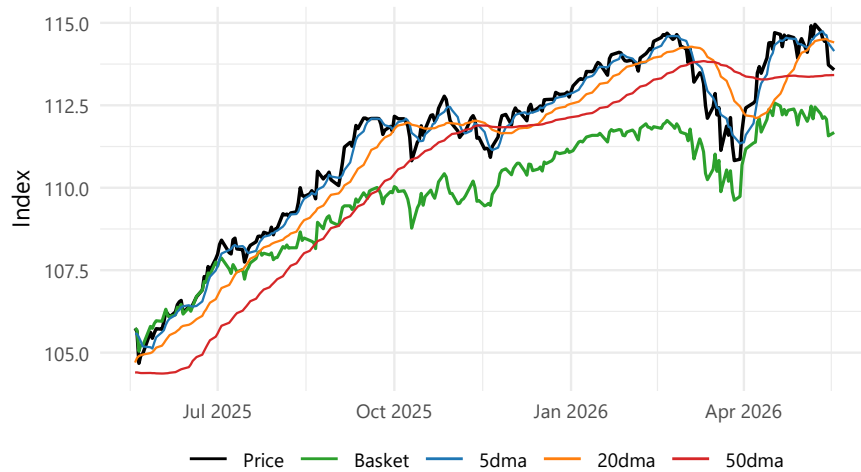
15	GNR	0.0548	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0541	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0528	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0527	iShares U.S. Basic Materials ETF	Materials	FLAT
21	NANR	0.0523	State Street SPDR S&P North American Na...	Materials	LONG
13	VAW	0.0513	Vanguard Materials ETF	Materials	FLAT
16	PICK	0.0508	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0505	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
1	XLB	0.0504	State Street Materials Select Sector SP...	Materials	FLAT
20	IGE	0.0501	iShares North American Natural Resource...	Energy	LONG
4	XME	0.0498	State Street SPDR S&P Metals & Mining E...	Materials	LONG
17	SLX	0.0495	VanEck Steel ETF	Materials	LONG
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0471	Global X Copper Miners ETF	Materials	LONG
5	EMLP	0.0442	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0441	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
18	NORW	0.0439	Global X MSCI Norway ETF	Norway Equity	LONG
7	AMLP	0.0419	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
8	MLPA	0.0407	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0364	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0332	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +7.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5155
Preferred Stock	7	0.3063
Emerging Markets Fixed Income	4	0.1783

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.050

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0466	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0463	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.763 **Signal:** LONG since 2026-05-06 (8 days, -2.1%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +15.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
VHT	1.103	Vanguard Health Care ETF	Healthcare	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	FLAT
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0572	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	FLAT
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0511	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0471	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – China Equity

Members: 15 **Var explained:** 0.804 **Signal:** LONG since 2026-04-15 (23 days, -1.4%)

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-02-27
- Current 252d return: +13.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-15 > last sell at 2026-02-27)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

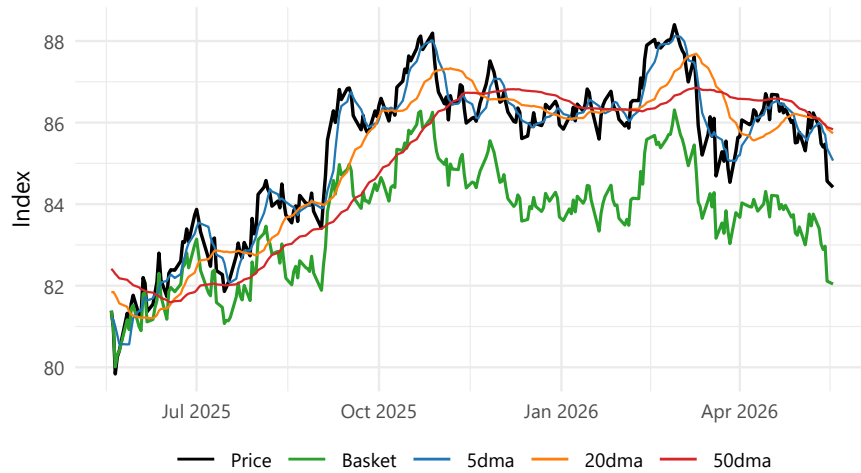
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0734	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	LONG
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	LONG
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	LONG
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0595	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
15	KURE	0.0588	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 13 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-10 – filter **PASS** (252d at date = +5.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +3.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-02-10)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4706
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
TLT	0.693	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

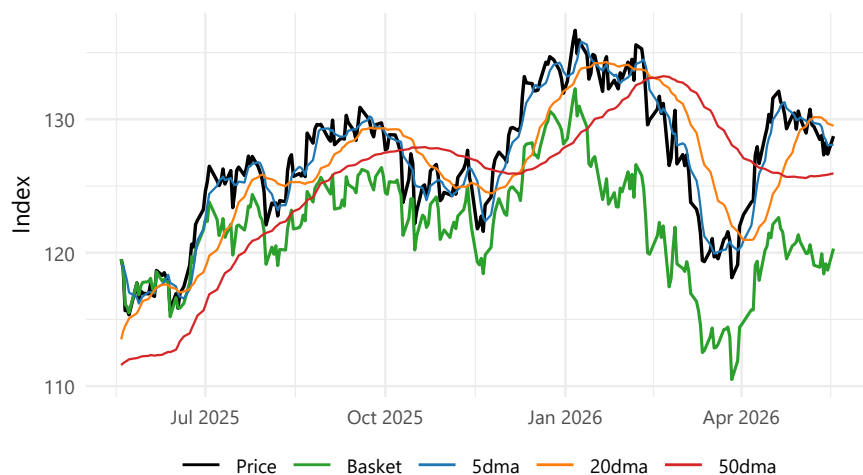
	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
11	USIG	0.0672	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
12	TLH	0.0667	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
1	TLT	0.0664	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
13	EDV	0.0657	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.878 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +8.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.938
Global Equity	1	0.062

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.016

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0754	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0754	Fidelity MSCI Financials Index ETF	Financials	LONG
5	FXO	0.0751	First Trust Financials AlphaDEX	Financials	FLAT
7	IYF	0.0751	iShares U.S. Financial ETF	Financials	FLAT
8	IYG	0.0746	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0740	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0732	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0711	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	FLAT
13	IAI	0.0699	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0641	State Street SPDR S&P Insurance ETF	Financials	FLAT
12	PSP	0.0620	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (26 days, 2.6%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-03-10
- Current 252d return: +10.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

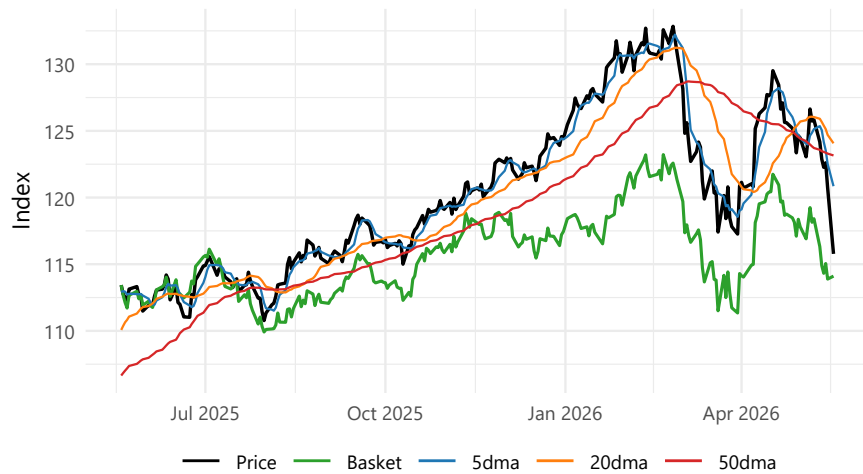
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – India Equity

Members: 14 **Var explained:** 0.361 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +15.0%)
- Latest sell_cross: 2026-05-12
- Current 252d return: +2.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-12 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.3276
Latin America Equity	1	0.1003
Brazil Equity	1	0.0891
Malaysia Equity	1	0.0884
Chile Equity	1	0.0825
Indonesia Equity	1	0.0812
Philippines Equity	1	0.0759
Vietnam Equity	1	0.0560
Turkey Equity	1	0.0438
Agricultural Commodities	1	0.0312
Platinum	1	0.0179
Natural Gas	1	0.0061

Replication basket – EWZ+INDA (cor=0.91) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
EWZ	0.254	iShares MSCI Brazil ETF	Brazil Equity	FLAT
INDA	0.507	iShares MSCI India ETF	India Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

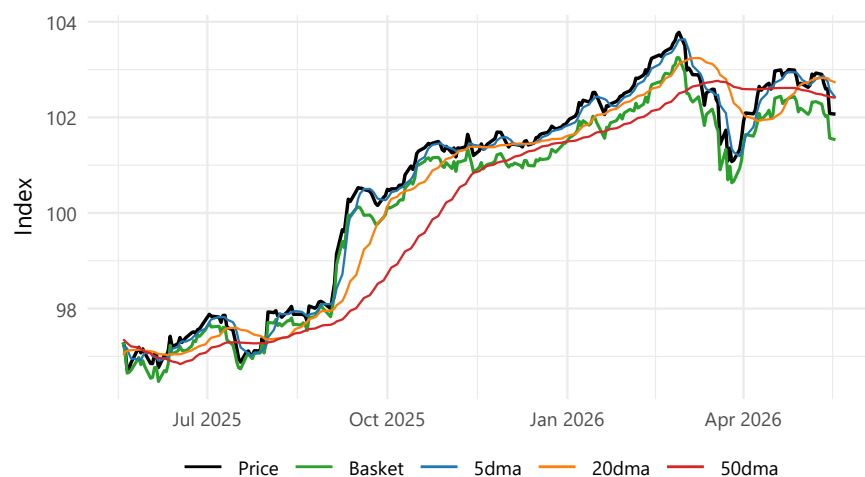
	ticker	PC_wt	name	category	etf_signal
2	INDA	0.1098	iShares MSCI India ETF	India Equity	FLAT
4	INDY	0.1093	iShares India 50 ETF	India Equity	FLAT
5	EPI	0.1086	WisdomTree India Earnings Fund	India Equity	FLAT
8	ILF	0.1003	iShares Latin America 40 ETF	Latin America Equity	FLAT
1	EWZ	0.0891	iShares MSCI Brazil ETF	Brazil Equity	FLAT
12	EWM	0.0884	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
7	ECH	0.0825	iShares MSCI Chile ETF	Chile Equity	LONG
10	EIDO	0.0812	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
11	EPHE	0.0759	iShares MSCI Philippines ETF	Philippines Equity	FLAT
14	VNM	0.0560	VanEck Vietnam ETF	Vietnam Equity	LONG
13	TUR	0.0438	iShares MSCI Turkey ETF	Turkey Equity	FLAT
3	DBA	0.0312	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
6	PPLT	0.0179	abrdn Physical Platinum Shares ETF	Platinum	LONG
9	UNG	0.0061	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 17 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-14 – filter **PASS** (252d at date = +6.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +4.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-14)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.181

ticker	rep_wt	name	category	etf_signal
MUB	0.905	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 18 – Gold

Members: 12 **Var explained:** 0.857 **Signal:** LONG since 2026-05-12 (4 days, -8.5%)

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-04-28
- Current 252d return: +67.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-12 > last sell at 2026-04-28)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1725
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
GLD	0.027	SPDR Gold Shares	Gold	FLAT
IAU	0.743	iShares Gold Trust Shares	Gold	FLAT
SLV	0.410	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0867	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	LONG
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	LONG
3	GLD	0.0835	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0828	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0817	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 19 – Consumer Staples

Members: 11 **Var explained:** 0.747 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-01 – filter **PASS** (252d at date = +9.8%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +7.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-01)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5727
Utilities	4	0.3526
U.S. Small Cap Consumer Staples	1	0.0746

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
XLU	0.365	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.587	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0974	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0947	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

3	KXI	0.0944	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0891	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0880	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0880	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0875	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	FLAT

Cluster 20 – Inflation-Protected Fixed Income

Members: 10 Var explained: 0.908 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +4.1%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP (cor=0.90) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
VTIP	1.724	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 21 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +39.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7798
Asia-Pacific Equity	2	0.2202

Replication basket – EWJ (cor=0.96) basket lead at lag 1: -0.036

ticker	rep_wt	name	category	etf_signal
EWJ	0.908	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1115	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1107	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT

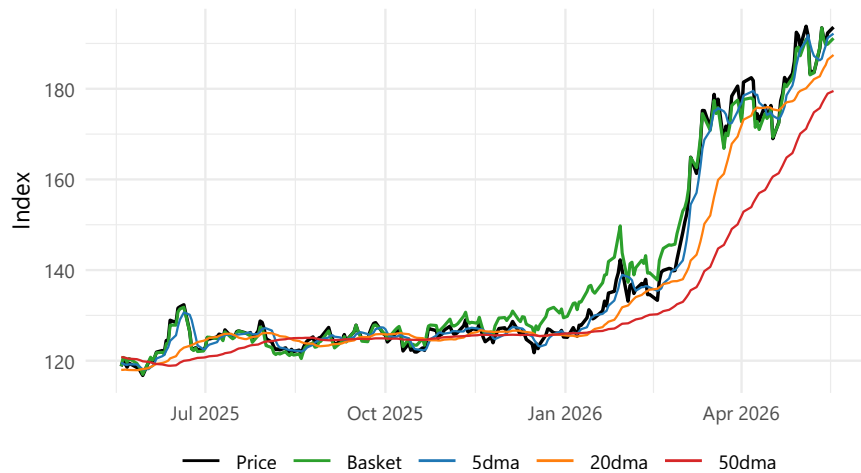
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1089	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 22 – Broad Commodities

Members: 9 **Var explained:** 0.912 **Signal:** LONG since 2026-01-08 (89 days, 51.4%)

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.7%)
- Latest sell_cross: 2025-12-15
- Current 252d return: +62.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-01-08 > last sell at 2025-12-15)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183

Energy Commodities	1	0.1119
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Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.071

ticker	rep_wt	name	category	etf_signal
PDBC	1.173	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

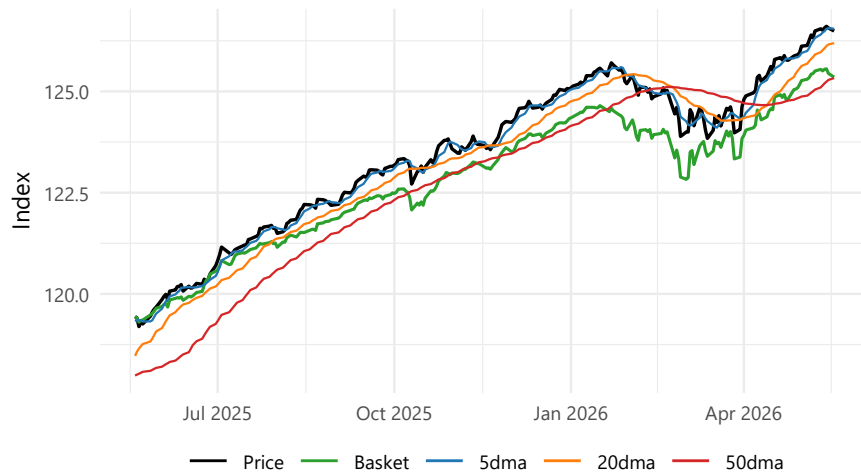
	ticker	PC_wt	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1149	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1148	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1098	United States Oil Fund	Oil	LONG
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1051	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1049	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (29 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-02-03
- Current 252d return: +6.0% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3631
Interest Rate Hedged Fixed Income	3	0.3246
Floating Rate Fixed Income	3	0.3123

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	rep_wt	name	category	etf_signal
FLOT	0.292	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

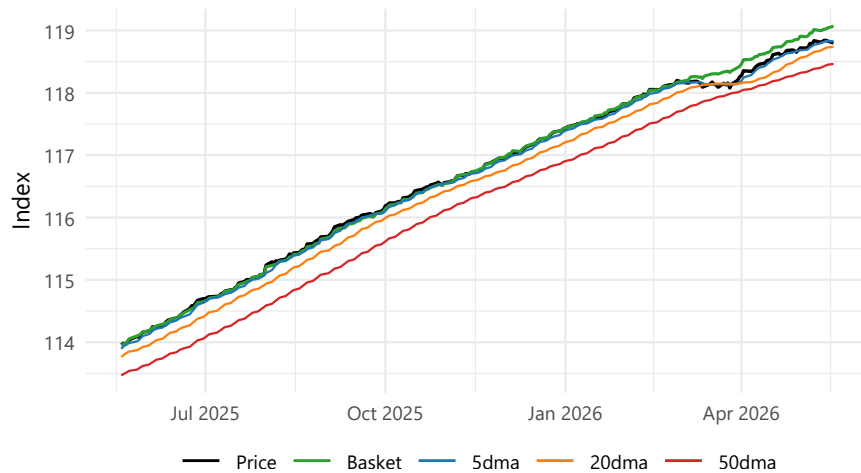
3	BKLN	0.1212	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1081	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1007	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.453 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-31 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.108

ticker	rep_wt	name	category	etf_signal
SHV	0.347	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.176	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.118	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.417	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

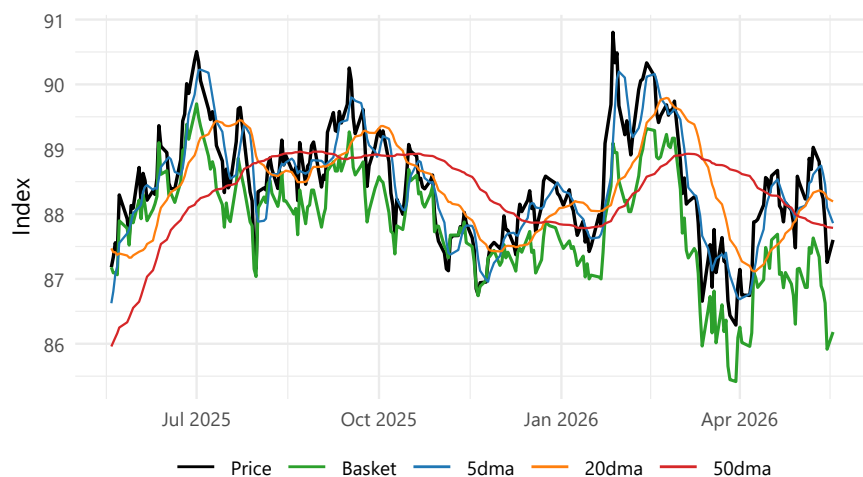
	ticker	PC_wt	name	category	etf_signal
5	GSY	0.1337	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1289	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1265	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1244	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1091	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1065	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.1034	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0884	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0791	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-30 – filter **PASS** (252d at date = +0.3%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +1.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-30)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.029

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1648	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXY	0.1452	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.624 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-04-08
- Current 252d return: -38.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
PTNQ	cluster_5	U.S. Large Cap Growth	2026-04-17	21	8.7%	0.00	1.00
PTLC	cluster_5	U.S. Large Cap Growth	2026-04-20	20	4.2%	0.00	1.00
IYW	cluster_5	U.S. Large Cap Growth	2026-04-10	26	21.7%	0.00	1.00
QQQ	cluster_5	U.S. Large Cap Growth	2026-04-10	26	15.5%	0.00	1.00
XLK	cluster_5	U.S. Large Cap Growth	2026-04-10	26	22.3%	0.00	1.00
FTEC	cluster_5	U.S. Large Cap Growth	2026-04-10	26	20.7%	0.00	1.00
VGT	cluster_5	U.S. Large Cap Growth	2026-04-10	26	20.7%	0.00	1.00
NXTG	cluster_5	U.S. Large Cap Growth	2026-04-08	28	18.8%	0.00	1.00
QTEC	cluster_5	U.S. Large Cap Growth	2026-04-10	26	27.7%	0.00	1.00
IXN	cluster_5	U.S. Large Cap Growth	2026-04-10	26	21.4%	0.00	1.00
IGM	cluster_5	U.S. Large Cap Growth	2026-04-09	27	20.2%	0.00	1.00
XT	cluster_5	U.S. Large Cap Growth	2026-04-13	25	10.2%	0.00	1.00
ONEQ	cluster_5	U.S. Large Cap Growth	2026-04-13	25	12.6%	0.00	1.00
XSD	cluster_5	U.S. Large Cap Growth	2026-04-09	27	46.3%	0.00	1.00
AIQ	cluster_5	U.S. Large Cap Growth	2026-04-13	25	20.0%	0.00	1.00
XNTK	cluster_5	U.S. Large Cap Growth	2026-04-10	26	20.5%	0.00	1.00
SMH	cluster_5	U.S. Large Cap Growth	2026-04-08	28	29.1%	0.00	1.00
IVW	cluster_5	U.S. Large Cap Growth	2026-04-13	25	11.2%	0.00	1.00
VOOG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	11.2%	0.00	1.00
SPYG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	11.2%	0.00	1.00
MGK	cluster_5	U.S. Large Cap Growth	2026-04-13	25	12.2%	0.00	1.00
VUG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	11.8%	0.00	1.00
FTXL	cluster_5	U.S. Large Cap Growth	2026-04-07	29	46.8%	0.00	1.00
IUSG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	10.8%	0.00	1.00
SOXX	cluster_5	U.S. Large Cap Growth	2026-04-08	28	33.9%	0.00	1.00
NULG	cluster_5	U.S. Large Cap Growth	2026-04-10	26	13.7%	0.00	1.00
QTUM	cluster_5	U.S. Large Cap Growth	2026-04-10	26	21.7%	0.00	1.00
IWY	cluster_5	U.S. Large Cap Growth	2026-04-14	24	8.5%	0.00	1.00
QQEW	cluster_5	U.S. Large Cap Growth	2026-04-16	22	7.3%	0.00	1.00
SCHG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	10.5%	0.00	1.00
PSI	cluster_5	U.S. Large Cap Growth	2026-04-07	29	43.6%	0.00	1.00
ILCG	cluster_5	U.S. Large Cap Growth	2026-04-13	25	11.0%	0.00	1.00
BOTZ	cluster_5	U.S. Large Cap Growth	2026-04-16	22	8.5%	0.00	1.00
PRNT	cluster_5	U.S. Large Cap Growth	2026-04-16	22	7.2%	0.00	1.00
ROBO	cluster_5	U.S. Large Cap Growth	2026-04-14	24	10.5%	0.00	1.00
VONG	cluster_5	U.S. Large Cap Growth	2026-04-14	24	7.8%	0.00	1.00
IWF	cluster_5	U.S. Large Cap Growth	2026-04-14	24	7.8%	0.00	1.00

PWB	cluster_5	U.S. Large Cap Growth	2026-04-10	26	11.0%	0.00	1.00
PSCT	cluster_5	U.S. Large Cap Growth	2026-04-07	29	24.6%	0.03	0.97
PTF	cluster_5	U.S. Large Cap Growth	2026-04-07	29	23.3%	0.05	0.95
TDIV	cluster_5	U.S. Large Cap Growth	2026-04-13	25	12.3%	0.08	0.92
FXL	cluster_5	U.S. Large Cap Growth	2026-04-09	27	14.1%	0.27	0.73
SPHB	cluster_5	U.S. Large Cap Growth	2026-04-10	26	10.9%	0.28	0.72
EIS	cluster_5	U.S. Large Cap Growth	2026-04-10	26	2.9%	0.60	0.40
QYLD	cluster_5	U.S. Large Cap Growth	2026-04-08	28	1.9%	0.83	0.17
BIL	cluster_24	Cash / Ultra-Short Fixe...	2022-05-19	1001	18.1%	0.00	1.00
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2022-08-18	939	17.5%	0.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixe...	2023-01-30	827	18.8%	0.00	1.00
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-04-02	31	0.5%	0.05	0.95
GSY	cluster_24	Cash / Ultra-Short Fixe...	2022-12-29	847	19.1%	0.16	0.84
LQDH	cluster_23	Bank Loan / Floating Ra...	2026-04-06	30	1.5%	0.00	1.00
IGHG	cluster_23	Bank Loan / Floating Ra...	2026-04-07	29	1.4%	0.00	1.00
IGBH	cluster_23	Bank Loan / Floating Ra...	2026-04-08	28	1.8%	0.00	1.00
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-04-07	29	1.7%	0.06	0.94
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-04-06	30	0.7%	0.45	0.55
FTSL	cluster_23	Bank Loan / Floating Ra...	2026-04-06	30	1.2%	0.55	0.45
DXJ	cluster_21	Japan Equity	2026-05-01	11	5.3%	0.00	1.00
EWJV	cluster_21	Japan Equity	2026-05-06	8	1.0%	0.00	1.00
HEWJ	cluster_21	Japan Equity	2026-04-08	28	2.4%	0.01	0.99
DBJP	cluster_21	Japan Equity	2026-04-08	28	2.2%	0.01	0.99
VPL	cluster_21	Japan Equity	2026-04-10	26	6.5%	0.02	0.98
BBJP	cluster_21	Japan Equity	2026-04-30	12	2.0%	0.02	0.98
EWJ	cluster_21	Japan Equity	2026-04-13	25	2.9%	0.03	0.97
PDBC	cluster_22	Broad Commodities	2025-12-22	100	41.1%	0.00	1.00
DBC	cluster_22	Broad Commodities	2025-12-22	100	40.2%	0.00	1.00
USO	cluster_22	Broad Commodities	2026-01-29	75	88.6%	0.04	0.96
XLG	cluster_4	U.S. Large Cap Equity	2026-04-13	25	10.5%	0.00	1.00
OEF	cluster_4	U.S. Large Cap Equity	2026-04-13	25	9.8%	0.00	1.00
DSI	cluster_4	U.S. Large Cap Equity	2026-04-14	24	8.5%	0.00	1.00
IOO	cluster_4	U.S. Large Cap Equity	2026-04-13	25	9.7%	0.00	1.00
MGC	cluster_4	U.S. Large Cap Equity	2026-04-13	25	8.8%	0.00	1.00
IWL	cluster_4	U.S. Large Cap Equity	2026-04-13	25	8.4%	0.00	1.00
ESGU	cluster_4	U.S. Large Cap Equity	2026-04-13	25	8.0%	0.00	1.00
SPY	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.7%	0.00	1.00
VOO	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.6%	0.00	1.00
VV	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.7%	0.00	1.00
IVV	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.6%	0.00	1.00
ILCB	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.6%	0.00	1.00
EPS	cluster_4	U.S. Large Cap Equity	2026-04-10	26	8.3%	0.00	1.00
OMFL	cluster_4	U.S. Large Cap Equity	2026-04-10	26	7.4%	0.00	1.00
SCHX	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.4%	0.00	1.00
SPYX	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.7%	0.00	1.00

SUSA	cluster_4	U.S. Large Cap Equity	2026-04-13	25	8.0%	0.00	1.00
SPTM	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.1%	0.00	1.00
IYY	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.2%	0.00	1.00
VONE	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.0%	0.00	1.00
IWB	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.0%	0.00	1.00
SCHB	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.1%	0.00	1.00
ITOT	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.1%	0.00	1.00
CATH	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.2%	0.00	1.00
VTI	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.0%	0.00	1.00
VTHR	cluster_4	U.S. Large Cap Equity	2026-04-13	25	7.0%	0.00	1.00
IWV	cluster_4	U.S. Large Cap Equity	2026-04-13	25	6.8%	0.00	1.00
QDF	cluster_4	U.S. Large Cap Equity	2026-04-13	25	6.1%	0.01	0.99
QUS	cluster_4	U.S. Large Cap Equity	2026-04-14	24	3.6%	0.01	0.99
LRGF	cluster_4	U.S. Large Cap Equity	2026-04-13	25	6.5%	0.01	0.99
SPHQ	cluster_4	U.S. Large Cap Equity	2026-04-10	26	4.9%	0.01	0.99
GSLC	cluster_4	U.S. Large Cap Equity	2026-04-13	25	6.3%	0.02	0.98
URTH	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.4%	0.03	0.97
DGRW	cluster_4	U.S. Large Cap Equity	2026-04-14	24	4.0%	0.04	0.96
ACWI	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.4%	0.05	0.95
JHML	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.1%	0.06	0.94
QAI	cluster_4	U.S. Large Cap Equity	2026-04-08	28	2.9%	0.07	0.93
TILT	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.5%	0.07	0.93
SPGM	cluster_4	U.S. Large Cap Equity	2026-04-10	26	5.9%	0.07	0.93
CRBN	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.5%	0.09	0.91
VT	cluster_4	U.S. Large Cap Equity	2026-04-13	25	5.0%	0.10	0.90
QUAL	cluster_4	U.S. Large Cap Equity	2026-04-14	24	3.8%	0.11	0.89
AOA	cluster_4	U.S. Large Cap Equity	2026-04-13	25	3.9%	0.15	0.85
EWS	cluster_4	U.S. Large Cap Equity	2026-04-07	29	2.9%	0.27	0.72
AOR	cluster_4	U.S. Large Cap Equity	2026-04-13	25	2.6%	0.30	0.70
AOM	cluster_4	U.S. Large Cap Equity	2026-04-13	25	1.4%	0.56	0.44
FPXI	cluster_4	U.S. Large Cap Equity	2026-04-10	26	5.2%	0.59	0.41
AOK	cluster_4	U.S. Large Cap Equity	2026-04-13	25	0.8%	0.75	0.25
XLC	cluster_4	U.S. Large Cap Equity	2026-04-14	24	0.4%	0.91	0.07
FCOM	cluster_4	U.S. Large Cap Equity	2026-04-13	25	3.6%	0.97	0.03
VOX	cluster_4	U.S. Large Cap Equity	2026-04-13	25	3.7%	0.97	0.03
IYC	cluster_4	U.S. Large Cap Equity	2026-04-14	24	-1.4%	1.00	-0.37
AMLX	cluster_9	Materials	2026-04-29	13	2.5%	0.00	1.00
MLPX	cluster_9	Materials	2026-04-30	12	1.5%	0.00	1.00
MLPA	cluster_9	Materials	2026-04-29	13	2.1%	0.01	0.99
SLX	cluster_9	Materials	2026-04-09	27	7.8%	0.03	0.97
DBB	cluster_9	Materials	2026-04-10	26	4.9%	0.03	0.97
NANR	cluster_9	Materials	2026-05-11	5	-1.9%	0.02	0.97
IGE	cluster_9	Materials	2026-05-14	2	0.7%	0.00	0.96
COPX	cluster_9	Materials	2026-05-08	6	-4.8%	0.09	0.91
NORW	cluster_9	Materials	2025-12-09	109	34.5%	0.19	0.81

EMLP	cluster_9	Materials	2026-05-14	2	-0.3%	0.25	0.50
XME	cluster_9	Materials	2026-04-15	23	-2.1%	0.78	0.21
CIBR	cluster_7	U.S. Mid Cap Growth	2026-04-17	21	21.7%	0.00	1.00
FV	cluster_7	U.S. Mid Cap Growth	2026-04-14	24	8.7%	0.00	1.00
HACK	cluster_7	U.S. Mid Cap Growth	2026-04-17	21	16.1%	0.00	1.00
MTUM	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	13.4%	0.00	1.00
BLCN	cluster_7	U.S. Mid Cap Growth	2026-04-15	23	8.1%	0.00	1.00
IGV	cluster_7	U.S. Mid Cap Growth	2026-04-21	19	7.1%	0.02	0.98
ICVT	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	7.5%	0.02	0.98
RPG	cluster_7	U.S. Mid Cap Growth	2026-04-09	27	9.7%	0.03	0.97
CWB	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	6.8%	0.05	0.95
PRN	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	9.1%	0.05	0.95
FTC	cluster_7	U.S. Mid Cap Growth	2026-04-10	26	5.3%	0.06	0.94
PDP	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	5.9%	0.07	0.93
XMMO	cluster_7	U.S. Mid Cap Growth	2026-04-06	30	9.1%	0.08	0.92
SKYY	cluster_7	U.S. Mid Cap Growth	2026-04-17	21	10.3%	0.12	0.88
ARKQ	cluster_7	U.S. Mid Cap Growth	2026-04-14	24	6.5%	0.13	0.87
VFMO	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	6.6%	0.22	0.78
ARKW	cluster_7	U.S. Mid Cap Growth	2026-04-14	24	9.2%	0.25	0.75
QMOM	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	3.9%	0.30	0.70
BLOK	cluster_7	U.S. Mid Cap Growth	2026-04-13	25	10.2%	0.30	0.70
FPX	cluster_7	U.S. Mid Cap Growth	2026-04-08	28	7.4%	0.30	0.70
IMCG	cluster_7	U.S. Mid Cap Growth	2026-04-09	27	6.4%	0.37	0.63
XAR	cluster_7	U.S. Mid Cap Growth	2026-05-12	4	-1.9%	0.25	0.36
DWAS	cluster_7	U.S. Mid Cap Growth	2026-04-07	29	8.8%	0.79	0.21
VOT	cluster_7	U.S. Mid Cap Growth	2026-04-14	24	2.7%	0.79	0.21
FDN	cluster_7	U.S. Mid Cap Growth	2026-04-10	26	11.3%	0.82	0.18
IWP	cluster_7	U.S. Mid Cap Growth	2026-04-15	23	0.5%	0.89	0.11
IPO	cluster_7	U.S. Mid Cap Growth	2026-04-13	25	6.0%	0.96	0.04
ARKK	cluster_7	U.S. Mid Cap Growth	2026-04-14	24	-0.4%	0.97	0.00
EWY	cluster_8	Emerging Markets Equity	2026-04-09	27	26.4%	0.00	1.00
AIA	cluster_8	Emerging Markets Equity	2026-04-10	26	15.6%	0.01	0.99
AAXJ	cluster_8	Emerging Markets Equity	2026-04-13	25	9.0%	0.09	0.91
EWT	cluster_8	Emerging Markets Equity	2026-04-08	28	20.2%	0.11	0.89
EEMA	cluster_8	Emerging Markets Equity	2026-04-13	25	8.5%	0.18	0.82
XSOE	cluster_8	Emerging Markets Equity	2026-04-10	26	7.9%	0.18	0.82
EMGF	cluster_8	Emerging Markets Equity	2026-04-10	26	8.0%	0.19	0.81
GEM	cluster_8	Emerging Markets Equity	2026-04-10	26	6.5%	0.22	0.78
EEM	cluster_8	Emerging Markets Equity	2026-04-10	26	7.3%	0.27	0.73
DEM	cluster_8	Emerging Markets Equity	2026-04-09	27	4.3%	0.27	0.73
EEMV	cluster_8	Emerging Markets Equity	2026-04-10	26	5.2%	0.31	0.69
ESGE	cluster_8	Emerging Markets Equity	2026-04-13	25	6.2%	0.33	0.67
IEMG	cluster_8	Emerging Markets Equity	2026-04-10	26	6.9%	0.36	0.64
GMF	cluster_8	Emerging Markets Equity	2026-04-14	24	2.7%	0.62	0.38
PXH	cluster_8	Emerging Markets Equity	2026-04-10	26	2.7%	0.68	0.32

EWX	cluster_8	Emerging Markets Equity	2026-04-10	26	4.7%	0.81	0.19
FNDE	cluster_8	Emerging Markets Equity	2026-04-10	26	1.7%	0.84	0.16
SPEM	cluster_8	Emerging Markets Equity	2026-04-13	25	2.5%	0.90	0.10
VWO	cluster_8	Emerging Markets Equity	2026-04-13	25	2.3%	0.93	0.07
SCHE	cluster_8	Emerging Markets Equity	2026-04-13	25	1.9%	0.93	0.07
DGS	cluster_8	Emerging Markets Equity	2026-04-10	26	1.4%	0.94	0.06
CNXT	cluster_12	China Equity	2026-04-10	26	14.2%	0.00	1.00
CQQQ	cluster_12	China Equity	2026-04-20	20	3.0%	0.01	0.99
EWI	cluster_12	China Equity	2026-04-08	28	0.8%	0.02	0.98
KBA	cluster_12	China Equity	2026-04-10	26	6.4%	0.03	0.97
ASHR	cluster_12	China Equity	2026-04-13	25	3.4%	0.05	0.95
CNYA	cluster_12	China Equity	2026-04-14	24	1.8%	0.07	0.93
CXSE	cluster_12	China Equity	2026-04-14	24	-0.3%	0.44	0.56
GXC	cluster_12	China Equity	2026-05-04	10	-1.2%	0.64	0.35
FXI	cluster_12	China Equity	2026-04-30	12	-1.7%	0.64	0.35
MCHI	cluster_12	China Equity	2026-05-06	8	-4.0%	0.78	0.04
KGRN	cluster_12	China Equity	2026-05-06	8	-3.7%	0.65	-0.51
VIDI	cluster_2	Developed ex-US Equity	2026-04-09	27	5.0%	0.02	0.98
FNDF	cluster_2	Developed ex-US Equity	2026-04-08	28	3.1%	0.04	0.96
PXF	cluster_2	Developed ex-US Equity	2026-04-08	28	3.1%	0.05	0.95
IDV	cluster_2	Developed ex-US Equity	2026-04-06	30	4.6%	0.08	0.92
HFXI	cluster_2	Developed ex-US Equity	2026-04-09	27	3.4%	0.16	0.84
IQDF	cluster_2	Developed ex-US Equity	2026-04-10	26	3.1%	0.22	0.78
SCHF	cluster_2	Developed ex-US Equity	2026-04-10	26	2.8%	0.29	0.71
SCZ	cluster_2	Developed ex-US Equity	2026-04-10	26	2.2%	0.30	0.70
SPDW	cluster_2	Developed ex-US Equity	2026-04-10	26	2.8%	0.32	0.68
EWN	cluster_2	Developed ex-US Equity	2026-04-10	26	2.9%	0.32	0.68
VEA	cluster_2	Developed ex-US Equity	2026-04-10	26	2.6%	0.33	0.67
IMTM	cluster_2	Developed ex-US Equity	2026-04-10	26	0.9%	0.39	0.61
EWI	cluster_2	Developed ex-US Equity	2026-05-05	9	0.4%	0.38	0.61
CWI	cluster_2	Developed ex-US Equity	2026-04-10	26	2.6%	0.42	0.58
VEU	cluster_2	Developed ex-US Equity	2026-04-10	26	2.8%	0.44	0.56
FEP	cluster_2	Developed ex-US Equity	2026-04-09	27	1.5%	0.45	0.55
IPKW	cluster_2	Developed ex-US Equity	2026-04-09	27	1.6%	0.45	0.55
DLS	cluster_2	Developed ex-US Equity	2026-04-13	25	0.4%	0.46	0.54
IXUS	cluster_2	Developed ex-US Equity	2026-04-10	26	2.8%	0.47	0.53
VXUS	cluster_2	Developed ex-US Equity	2026-04-10	26	2.8%	0.47	0.53
DWX	cluster_2	Developed ex-US Equity	2026-05-04	10	1.4%	0.46	0.52
ACWX	cluster_2	Developed ex-US Equity	2026-04-10	26	2.5%	0.50	0.50
EWI	cluster_2	Developed ex-US Equity	2026-04-08	28	3.1%	0.54	0.46
VIGI	cluster_2	Developed ex-US Equity	2026-04-14	24	0.6%	0.54	0.46
GWX	cluster_2	Developed ex-US Equity	2026-04-13	25	2.2%	0.55	0.45
VYMI	cluster_2	Developed ex-US Equity	2026-04-09	27	1.1%	0.57	0.43
EWL	cluster_2	Developed ex-US Equity	2026-04-15	23	0.4%	0.61	0.39
VSS	cluster_2	Developed ex-US Equity	2026-04-13	25	1.8%	0.63	0.37

ENZL	cluster_2	Developed ex-US Equity	2026-05-06	8	-4.3%	0.57	0.36
FNDC	cluster_2	Developed ex-US Equity	2026-04-10	26	1.4%	0.65	0.35
PID	cluster_2	Developed ex-US Equity	2026-04-30	12	0.2%	0.67	0.32
HDEF	cluster_2	Developed ex-US Equity	2026-04-07	29	1.9%	0.66	0.29
SCHC	cluster_2	Developed ex-US Equity	2026-04-13	25	0.7%	0.72	0.28
FXA	cluster_2	Developed ex-US Equity	2026-04-13	25	1.1%	0.80	0.20
IHDG	cluster_2	Developed ex-US Equity	2026-05-06	8	-1.9%	0.73	0.15
EWO	cluster_2	Developed ex-US Equity	2026-04-10	26	2.3%	0.86	0.14
BBAX	cluster_2	Developed ex-US Equity	2026-04-30	12	-1.0%	0.75	0.10
EWC	cluster_2	Developed ex-US Equity	2026-04-09	27	2.0%	0.98	0.02
EWG	cluster_2	Developed ex-US Equity	2026-04-14	24	0.4%	0.97	0.02
NFRA	cluster_2	Developed ex-US Equity	2026-04-09	27	-0.2%	0.97	0.01
EZA	cluster_2	Developed ex-US Equity	2026-05-08	6	-5.4%	0.78	-0.30
EWD	cluster_2	Developed ex-US Equity	2026-05-06	8	-4.1%	1.00	-0.95
SIVR	cluster_18	Gold	2026-05-11	5	-10.3%	0.05	0.94
SLV	cluster_18	Gold	2026-05-11	5	-10.3%	0.05	0.94
SILJ	cluster_18	Gold	2026-05-11	5	-12.4%	0.15	0.79
SIL	cluster_18	Gold	2026-05-11	5	-11.1%	0.22	0.68
GLTR	cluster_18	Gold	2026-05-12	4	-6.3%	0.29	0.51
GDJX	cluster_18	Gold	2026-05-12	4	-11.1%	0.46	0.14
DBP	cluster_18	Gold	2026-05-12	4	-5.1%	0.44	0.14
GOAU	cluster_18	Gold	2026-05-13	3	-8.8%	0.44	0.01
GDJX	cluster_18	Gold	2026-05-12	4	-10.3%	0.51	-0.09
VLUE	cluster_1	U.S. Large Cap Value	2026-04-09	27	16.5%	0.00	1.00
IWX	cluster_1	U.S. Large Cap Value	2026-04-10	26	6.0%	0.00	1.00
MGV	cluster_1	U.S. Large Cap Value	2026-04-13	25	3.4%	0.00	1.00
FNDC	cluster_1	U.S. Large Cap Value	2026-04-09	27	5.5%	0.00	1.00
RWL	cluster_1	U.S. Large Cap Value	2026-04-10	26	5.7%	0.00	1.00
PRF	cluster_1	U.S. Large Cap Value	2026-04-09	27	5.9%	0.00	1.00
ILCV	cluster_1	U.S. Large Cap Value	2026-04-13	25	4.4%	0.00	1.00
HDV	cluster_1	U.S. Large Cap Value	2026-05-01	11	0.9%	0.01	0.99
VTV	cluster_1	U.S. Large Cap Value	2026-04-13	25	2.4%	0.01	0.99
SCHD	cluster_1	U.S. Large Cap Value	2026-04-21	19	3.4%	0.01	0.99
VONV	cluster_1	U.S. Large Cap Value	2026-04-10	26	4.9%	0.02	0.98
IWD	cluster_1	U.S. Large Cap Value	2026-04-10	26	4.9%	0.02	0.98
DLN	cluster_1	U.S. Large Cap Value	2026-04-13	25	3.5%	0.03	0.97
PWV	cluster_1	U.S. Large Cap Value	2026-04-02	31	5.4%	0.04	0.96
SCHV	cluster_1	U.S. Large Cap Value	2026-04-10	26	3.6%	0.04	0.96
VOOV	cluster_1	U.S. Large Cap Value	2026-04-14	24	3.4%	0.06	0.94
RPV	cluster_1	U.S. Large Cap Value	2026-04-20	20	1.2%	0.06	0.94
IVE	cluster_1	U.S. Large Cap Value	2026-04-14	24	3.4%	0.07	0.93
SPYV	cluster_1	U.S. Large Cap Value	2026-04-14	24	3.4%	0.07	0.93
USMV	cluster_1	U.S. Large Cap Value	2026-05-05	9	1.6%	0.09	0.91
DTD	cluster_1	U.S. Large Cap Value	2026-04-13	25	3.2%	0.11	0.89
IUSV	cluster_1	U.S. Large Cap Value	2026-04-14	24	3.2%	0.19	0.81

NULV	cluster_1	U.S. Large Cap Value	2026-04-10	26	6.4%	0.19	0.81
JVAL	cluster_1	U.S. Large Cap Value	2026-04-10	26	7.3%	0.20	0.80
DHS	cluster_1	U.S. Large Cap Value	2026-04-20	20	1.2%	0.21	0.79
PFM	cluster_1	U.S. Large Cap Value	2026-04-14	24	2.8%	0.31	0.69
VIG	cluster_1	U.S. Large Cap Value	2026-04-14	24	2.9%	0.36	0.64
ACWV	cluster_1	U.S. Large Cap Value	2026-05-06	8	0.7%	0.41	0.57
FDVV	cluster_1	U.S. Large Cap Value	2026-04-14	24	2.9%	0.46	0.54
DGRO	cluster_1	U.S. Large Cap Value	2026-04-13	25	2.1%	0.49	0.51
RDIV	cluster_1	U.S. Large Cap Value	2026-04-21	19	1.0%	0.54	0.46
OUSA	cluster_1	U.S. Large Cap Value	2026-04-20	20	1.0%	0.56	0.44
SPYD	cluster_1	U.S. Large Cap Value	2026-04-21	19	1.3%	0.58	0.42
AIVL	cluster_1	U.S. Large Cap Value	2026-04-10	26	1.6%	0.58	0.42
DIA	cluster_1	U.S. Large Cap Value	2026-04-14	24	2.4%	0.59	0.41
FEX	cluster_1	U.S. Large Cap Value	2026-04-09	27	2.3%	0.60	0.40
SDOG	cluster_1	U.S. Large Cap Value	2026-04-07	29	2.1%	0.64	0.36
DIV	cluster_1	U.S. Large Cap Value	2026-04-07	29	2.0%	0.66	0.33
VYM	cluster_1	U.S. Large Cap Value	2026-04-10	26	3.0%	0.67	0.33
IMCB	cluster_1	U.S. Large Cap Value	2026-04-10	26	4.0%	0.78	0.22
SIZE	cluster_1	U.S. Large Cap Value	2026-04-13	25	2.4%	0.85	0.15
XLI	cluster_1	U.S. Large Cap Value	2026-04-13	25	-1.1%	0.86	0.14
FIDU	cluster_1	U.S. Large Cap Value	2026-04-13	25	-0.7%	0.87	0.13
VIS	cluster_1	U.S. Large Cap Value	2026-04-13	25	-0.7%	0.87	0.13
IWR	cluster_1	U.S. Large Cap Value	2026-04-10	26	3.0%	0.89	0.11
IWS	cluster_1	U.S. Large Cap Value	2026-04-09	27	2.1%	0.89	0.11
RSP	cluster_1	U.S. Large Cap Value	2026-04-14	24	1.6%	0.92	0.08
VO	cluster_1	U.S. Large Cap Value	2026-04-13	25	2.2%	0.96	0.04
PAVE	cluster_1	U.S. Large Cap Value	2026-04-10	26	0.1%	0.98	0.02
ONEY	cluster_1	U.S. Large Cap Value	2026-04-10	26	2.1%	0.99	0.01
IYJ	cluster_1	U.S. Large Cap Value	2026-04-13	25	-1.3%	0.99	0.01
SPHD	cluster_1	U.S. Large Cap Value	2026-05-18	0	0.0%	0.89	-0.03
SMMV	cluster_1	U.S. Large Cap Value	2026-04-09	27	-1.2%	1.00	-0.98
FIW	cluster_1	U.S. Large Cap Value	2026-05-08	6	-4.3%	1.00	-1.00
REZ	cluster_15	Real Estate	2026-04-14	24	2.7%	0.14	0.86
NURE	cluster_15	Real Estate	2026-04-13	25	4.5%	0.75	0.25
RWR	cluster_15	Real Estate	2026-04-10	26	3.3%	0.86	0.14
USRT	cluster_15	Real Estate	2026-04-09	27	3.9%	0.88	0.12
PSR	cluster_15	Real Estate	2026-04-09	27	3.8%	0.94	0.06
SCHH	cluster_15	Real Estate	2026-04-09	27	3.1%	0.95	0.05
REET	cluster_15	Real Estate	2026-04-13	25	1.9%	0.98	0.02
RWO	cluster_15	Real Estate	2026-04-13	25	1.3%	0.99	0.01
ICF	cluster_15	Real Estate	2026-04-08	28	3.9%	0.99	0.01
XLRE	cluster_15	Real Estate	2026-04-09	27	2.4%	0.99	0.01
FREL	cluster_15	Real Estate	2026-04-10	26	2.2%	0.99	0.01
VNQ	cluster_15	Real Estate	2026-04-10	26	2.2%	1.00	0.00
IYR	cluster_15	Real Estate	2026-04-10	26	1.9%	1.00	0.00

IVOG	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.7%	0.44	0.56
IJK	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.8%	0.48	0.52
MDYG	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.8%	0.49	0.51
IWC	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	4.9%	0.62	0.38
VBK	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	6.0%	0.69	0.31
VTWG	cluster_3	U.S. Mid Cap Equity	2026-04-13	25	5.5%	0.70	0.30
IWO	cluster_3	U.S. Mid Cap Equity	2026-04-13	25	5.5%	0.70	0.30
SCHM	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	4.0%	0.73	0.27
FTXR	cluster_3	U.S. Mid Cap Equity	2026-04-17	21	-1.3%	0.74	0.26
SCHA	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	4.9%	0.76	0.24
XMHQ	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	1.7%	0.79	0.20
IWM	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	5.6%	0.84	0.16
VTWO	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	5.6%	0.84	0.16
ISCG	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	4.1%	0.89	0.11
OMFS	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.9%	0.91	0.09
PSC	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	5.4%	0.93	0.07
ISMD	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	5.7%	0.93	0.07
SMLF	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.8%	0.94	0.06
IJH	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.4%	0.95	0.05
MDY	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.4%	0.95	0.05
IVOO	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.4%	0.95	0.05
SPMD	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.4%	0.95	0.05
IWN	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.2%	0.95	0.05
VTWV	cluster_3	U.S. Mid Cap Equity	2026-04-09	27	3.2%	0.95	0.05
VB	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	3.4%	0.96	0.04
VXF	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	4.7%	0.97	0.03
JHMM	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.1%	0.97	0.03
XSMO	cluster_3	U.S. Mid Cap Equity	2026-04-08	28	3.4%	1.00	0.00
IJS	cluster_3	U.S. Mid Cap Equity	2026-04-10	26	2.7%	1.00	0.00
JETS	cluster_3	U.S. Mid Cap Equity	2026-05-06	8	-6.2%	0.84	-0.15
XHB	cluster_3	U.S. Mid Cap Equity	2026-05-06	8	-8.8%	1.00	-1.00
FXD	cluster_3	U.S. Mid Cap Equity	2026-05-06	8	-5.4%	1.00	-1.00
PFXF	cluster_10	High Yield Bond	2026-04-14	24	1.0%	0.52	0.48
HYEM	cluster_10	High Yield Bond	2026-04-13	25	0.4%	0.87	0.13
PFF	cluster_10	High Yield Bond	2026-04-15	23	-0.2%	0.89	0.11
VRP	cluster_10	High Yield Bond	2026-04-10	26	-0.2%	0.93	0.07
HYLS	cluster_10	High Yield Bond	2026-04-09	27	0.1%	1.00	0.00
IHF	cluster_11	Healthcare	2026-05-08	6	2.0%	0.00	1.00
FXH	cluster_11	Healthcare	2026-04-10	26	1.3%	0.66	0.34
PSCH	cluster_11	Healthcare	2026-04-13	25	3.8%	0.98	0.02
IBB	cluster_11	Healthcare	2026-05-06	8	-5.7%	0.87	-0.57
IAI	cluster_14	Financials	2026-04-10	26	5.3%	0.73	0.27
KCE	cluster_14	Financials	2026-04-13	25	3.9%	0.91	0.09
XLF	cluster_14	Financials	2026-04-13	25	0.2%	1.00	-0.01
FNCL	cluster_14	Financials	2026-04-13	25	-0.0%	1.00	-0.01

TDDT	cluster_20	Inflation-Protected Fix...	2026-04-01	32	0.9%	0.78	0.22
STPZ	cluster_20	Inflation-Protected Fix...	2026-04-02	31	0.6%	0.82	0.18
TDTF	cluster_20	Inflation-Protected Fix...	2026-04-09	27	0.3%	1.00	0.00
TIPX	cluster_20	Inflation-Protected Fix...	2026-04-07	29	0.3%	1.00	0.00
IYK	cluster_19	Consumer Staples	2026-05-06	8	2.0%	0.00	1.00
XLP	cluster_19	Consumer Staples	2026-05-05	9	2.2%	0.00	1.00
KXI	cluster_19	Consumer Staples	2026-05-06	8	1.0%	0.00	1.00
VDC	cluster_19	Consumer Staples	2026-05-01	11	1.1%	0.00	1.00
FSTA	cluster_19	Consumer Staples	2026-05-01	11	1.1%	0.00	1.00
FXF	cluster_25	International Fixed Inc...	2026-05-05	9	-0.2%	0.88	0.02
SUB	cluster_17	Municipal Bond	2026-05-11	5	-0.2%	1.00	-0.99
SPSB	cluster_6	Aggregate Fixed Income	2026-04-09	27	-0.0%	1.00	0.00
AGZ	cluster_6	Aggregate Fixed Income	2026-05-11	5	-0.5%	1.00	-1.00
EWM	cluster_16	India Equity	2026-04-22	18	0.9%	0.11	0.89
VNM	cluster_16	India Equity	2026-04-13	25	5.5%	0.71	0.29
PPLT	cluster_16	India Equity	2026-05-11	5	-90.8%	0.98	-0.96
ECH	cluster_16	India Equity	2026-05-06	8	-5.3%	1.00	-1.00
UUP	cluster_26	Leveraged / Inverse	2026-05-18	0	0.0%	0.00	0.84

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
CIBR	cluster_7	U.S. Mid Cap Growth	1.00	1.00
IHF	cluster_11	Healthcare	1.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
FLOT	cluster_23	Bank Loan / Floating Rate	1.00	1.00
PTNQ	cluster_5	U.S. Large Cap Growth	1.00	1.00
IBDR	cluster_6	Aggregate Fixed Income	1.00	1.00
XLG	cluster_4	U.S. Large Cap Equity	1.00	1.00
IYK	cluster_19	Consumer Staples	1.00	1.00
VLUE	cluster_1	U.S. Large Cap Value	1.00	1.00
FTGC	cluster_22	Broad Commodities	1.00	1.00
AMLPG	cluster_9	Materials	1.00	1.00
DXJ	cluster_21	Japan Equity	1.00	1.00
CNXT	cluster_12	China Equity	1.00	1.00
EWY	cluster_8	Emerging Markets Equity	1.00	1.00
DBA	cluster_16	India Equity	1.00	0.99
VIDI	cluster_2	Developed ex-US Equity	1.00	0.98
SIVR	cluster_18	Gold	0.99	0.94
REZ	cluster_15	Real Estate	1.00	0.86
VTIP	cluster_20	Inflation-Protected Fixed...	1.00	0.84
IVOG	cluster_3	U.S. Mid Cap Equity	1.00	0.56

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
RSPD	cluster_3	U.S. Mid Cap Equity	1.00	-1.00
EWZ	cluster_16	India Equity	1.00	-1.00
PHO	cluster_1	U.S. Large Cap Value	1.00	-1.00
IEI	cluster_6	Aggregate Fixed Income	1.00	-1.00
TLT	cluster_13	Fixed Income	1.00	-1.00
XHE	cluster_11	Healthcare	1.00	-1.00

FUTY	cluster_19	Consumer Staples	1.00	-1.00
IBUY	cluster_7	U.S. Mid Cap Growth	1.00	-1.00
SQQQ	cluster_26	Leveraged / Inverse	1.00	-1.00
SHM	cluster_17	Municipal Bond	1.00	-1.00
ESPO	cluster_4	U.S. Large Cap Equity	1.00	-1.00
LTPZ	cluster_20	Inflation-Protected Fixed...	1.00	-1.00
EMQQ	cluster_12	China Equity	1.00	-0.99
RWX	cluster_2	Developed ex-US Equity	1.00	-0.96
DVYE	cluster_8	Emerging Markets Equity	1.00	-0.89
MOO	cluster_9	Materials	0.89	-0.88
BWX	cluster_25	International Fixed Income	1.00	-0.84
SGOL	cluster_18	Gold	0.83	-0.77
IAT	cluster_14	Financials	1.00	-0.76
PGF	cluster_10	High Yield Bond	1.00	-0.75

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_16	14	LONG	0.407	0.134	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_16	14	FLAT	0.491	0.080	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.692	0.040	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.732	0.095	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.628	0.221	0.00035	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_16	14	FLAT	0.595	0.173	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.779	0.131	0.00031	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.825	0.021	0.00000	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.785	0.164	0.00038	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.825	0.089	0.00022	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.647	0.187	0.00047	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.719	0.186	0.00042	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.785	0.180	0.00042	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_16	14	FLAT	0.489	0.220	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_6	43	FLAT	0.844	0.125	0.00034	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_6	43	FLAT	0.876	0.161	0.00030	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	LONG	0.819	0.203	0.00048	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_6	43	FLAT	0.837	0.233	0.00045	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_17	13	FLAT	0.877	0.250	0.00044	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_16	14	LONG	0.454	0.332	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_17	13	LONG	0.877	0.263	0.00052	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_12	15	FLAT	0.804	0.269	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_9	21	LONG	0.752	0.356	0.00083	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_16	14	FLAT	0.510	0.373	0.00106	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_12	15	LONG	0.856	0.255	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_16	14	FLAT	0.580	0.408	0.00107	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.864	0.332	0.00071	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_12	15	LONG	0.904	0.240	0.00063	
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_16	14	LONG	0.638	0.430	0.00118	
FLRN	State Street SPDR Bloomberg Inv...	Floating Rate Fixed I...	cluster_23	9	FLAT	0.904	0.320	0.00067	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20